Mid-Term Project Report
Privacy-Preserving Digital KYC System Using Zero-Knowledge Proofs

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1. Project Overview

The project aims to develop a privacy-preserving Digital KYC verification system using Zero-Knowledge Proofs (ZKPs). The objective is to allow a user to prove required KYC attributes, such as "KYC verified" or "age $\geq$ 18", without revealing the underlying sensitive personal information. The proposed architecture consists of a KYC Authority, a user-controlled digital credential and wallet, a ZK proof generation component, and a Service Provider that verifies the proof without accessing the user's raw personal data.

The project focuses on data minimization, selective disclosure, credential integrity, privacy-preserving verification, and practical evaluation of the cryptographic system.

2. Problem Motivation

Traditional KYC processes often require users to submit complete identity documents even when a service provider needs only one attribute. This can unnecessarily expose information such as date of birth, address, and government identification details. Centralized storage also increases privacy and security risks.

Our approach changes verification from "revealing data" to "proving a claim". The user retains the credential and generates a zero-knowledge proof for the requested claim, while the Service Provider receives only the proof and verification result.

3. Proposed Approach

The proposed workflow is:

KYC Authority → Digital Credential → User Wallet → ZK Proof → Service Provider

The KYC Authority verifies identity and issues a digital credential. The credential is stored locally by the user. When verification is required, the wallet generates a proof for the requested attribute. The Service Provider verifies the proof without receiving the underlying personal information.

The initial cryptographic claims are: **(i) age** ≥ 18 and **(ii) KYC status = verified**. (If time permits, the team may also explore the use of blockchain technology as an extension to the proposed system.)

4. Work Completed So Far

The group has completed the literature survey and presentation covering privacy-preserving credentials, selective disclosure, ZKPs, decentralized identity, credential revocation, range and membership proofs, and practical ZKP implementations.

The survey examined ten relevant papers and systems. Key ideas identified for implementation include cryptographic commitments, selective attribute disclosure, client-side ZK proofs, Merkle-based credential representations, range and membership proofs, commit-and-prove optimization, and practical proving and verification performance.

The implementation roadmap is divided into three stages: ZKP design, credential and wallet development, and proof verification with performance evaluation.

5. Individual Contributions

Each member has a primary cryptographic responsibility, while all members participate in integration, testing, evaluation, and documentation. The allocation is designed so that the four members contribute to different stages of the same cryptographic pipeline.

Member	Literature Survey Contribution	Primary Project Contribution
Reddi Pallavi	Selective disclosure of Verifiable Credentials, Crescent, and Zident; focused on commitments, ZKPs, privacy-preserving attribute proofs, and unlinkable presentations.	**Credential Cryptography and Selective Disclosure:** design the digital credential, study hashing and commitments, define private/public attributes, implement credential integrity, and design how only required attributes are disclosed. Contribute to privacy analysis and credential-to-ZKP integration.
B. Vaishnavi	Scalable DID and verifiable data sharing, ZKP-based age verification, and zk-X509; focused on digital identity, age proofs, credential revocation, and privacy-preserving verification.	**ZKP Circuit and Claim Design:** design Circom circuits for age $\geq$ 18 and KYC status, define public/private signals, implement constraints, test valid and invalid inputs, and optimize circuit complexity. Contribute to credential-to- circuit integration.
Durbasmriti Saha	ZKVault and Merkle Tree–Groth16 eKYC; focused on Circom, SnarkJS, Groth16, Merkle-based attribute hashing, proof generation, and verification performance.	**Groth16 Proving and Cryptographic Pipeline:** study the proving workflow, configure the proving environment, generate witnesses and proofs, integrate the prover with the credential/wallet flow, and evaluate proof-generation time and proof size.
Someshwar Singh	On-chain KYC permissioning and zkFaith; focused on range, equality and membership proofs, commit-and-prove optimization, dynamic credentials, and security properties.	**Verification, Security and Optimization:** implement and test proof verification, analyze soundness and privacy properties, test invalid/tampered proofs and incorrect inputs, study membership mechanisms where applicable, and evaluate verification time and scalability.

6. Development Timeline

The work is designed to proceed in parallel during the initial development period, followed by scheduled integration and evaluation.

Week	Planned Work
1	All members: ZKP/Groth16 fundamentals, system architecture, requirements, public/private input definition, and development setup.
2	Parallel: Pallavi–credential/commitment design; Vaishnavi–ZKP circuit design; Durba–Groth16 proving setup; Someshwar–verification and security design.
3	Parallel implementation of the four cryptographic modules: credential, circuits, prover, and verifier.
4	**Integration 1:** connect credential, circuit, prover and verifier; demonstrate age $\geq$ 18 proof without revealing DOB.
5	**Integration 2:** add KYC = verified and test valid, underage, unverified, and tampered-credential cases.
6	Integrate wallet, proof-generation service, and Service Provider verification interface; finalize the end-to-end prototype.
7	Performance, privacy and security evaluation: proof-generation time, proof size, verification time, constraint count, correctness, and scalability.
8	Final testing, documentation, results analysis, presentation, and demonstration preparation.

7. Shared Responsibilities and Evaluation

Although each member owns a primary cryptographic area, the implementation is collaborative. All four members will participate in cryptographic design discussions, integration, testing, security analysis, performance evaluation, and final documentation.

The system will be evaluated using proof-generation time, proof size, verification time, constraint complexity, correctness, privacy, and scalability. Security testing will include invalid proofs, incorrect public inputs, modified credentials, and unauthorized changes.

8. Expected Outcome

The expected outcome is a functional prototype demonstrating privacy-preserving KYC verification using digital credentials and Zero-Knowledge Proofs. The system should allow a user to prove selected KYC attributes while keeping unnecessary personal information private. If time permits, the team may further explore blockchain integration as an extension to the core ZKP-based KYC system.